Name: ____________ Date: ____________ Class: ____________

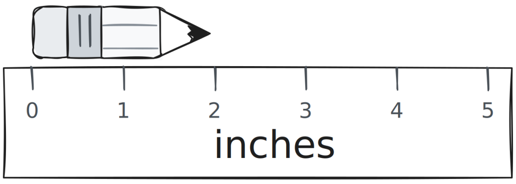
Lesson 1: Observations Worksheet — G7 Physics

Instructions: Answer all questions. Write in complete sentences unless told otherwise, and show your work for calculations.

1. In your own words, what is an **observation**?

2. Explain the difference between a **qualitative** statement and a **quantitative** statement, and give one example of each.

3. Explain why **“this pencil is small”** is **qualitative**, but **“this pencil measures 2 in”** is **quantitative**.



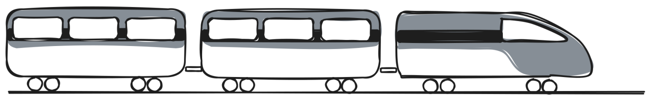
4. Explain the difference between a **subjective** statement and an **objective** statement.

5. Classify each statement below as **subjective** or **objective**, and justify your choice.
   (a) “The soup tastes salty.” → ________________________
   (b) “The soup contains 800 mg of sodium.” → ________________________
   (c) “Pikachu is cute.” → ________________________
   (d) “Pikachu is yellow.” → ________________________
6. Write **one statement** that is both qualitative and objective, and explain why it fits both.

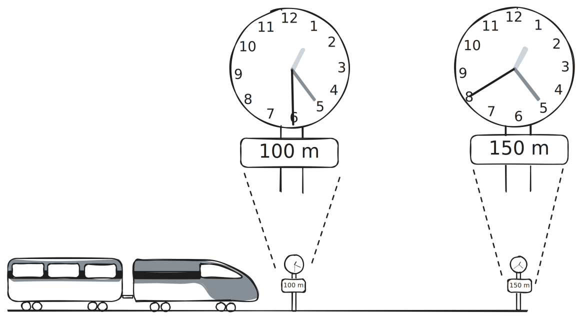
7. List the **three requirements** of a scientific statement, and explain one of them in your own words.

8. Classify each statement as **scientific** or **non-scientific**, and justify your choice.
   (a) “The rock has a mass of 3 kg.” → ________________________
   (b) “The music is too loud.” → ________________________

9. Rewrite **“this train is fast”** as a **testable scientific statement**, and state whether it is **falsifiable**.



10. The diagram shows a setup for measuring the speed of a toy train. Explain how you would use this setup to measure the train’s speed, using the words **observation**, **measurement**, and **testable** in your answer. Then write an **“if. . . then. . . ” hypothesis** for it.



Exit Ticket